MILITARY SPECIFICATION SHEET

CAPACITORS, VARIABLE, VACUUM OR GAS DIELECTRIC

STYLES CG65 AND CG66

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of MIL-C-23183.

GLASS ENVELOPE UNIT INACTIVE FOR NEW EQUIPMENT DESIGN AFTER 7 APRIL 1972

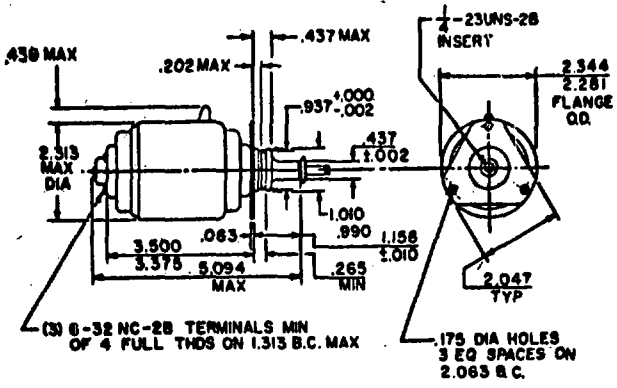


FIGURE 1a. Capacitor assembly for CG65E050-471G and CG66E050-471G.

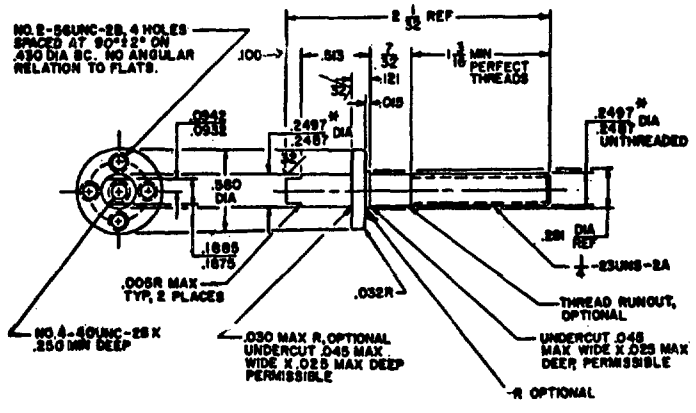


FIGURE 1b. Lead screw detail for CG65E050-471G.

FIGURE 1. Style CG65 and CG66 capacitors.

(B) denotes changes.

FSC 5910

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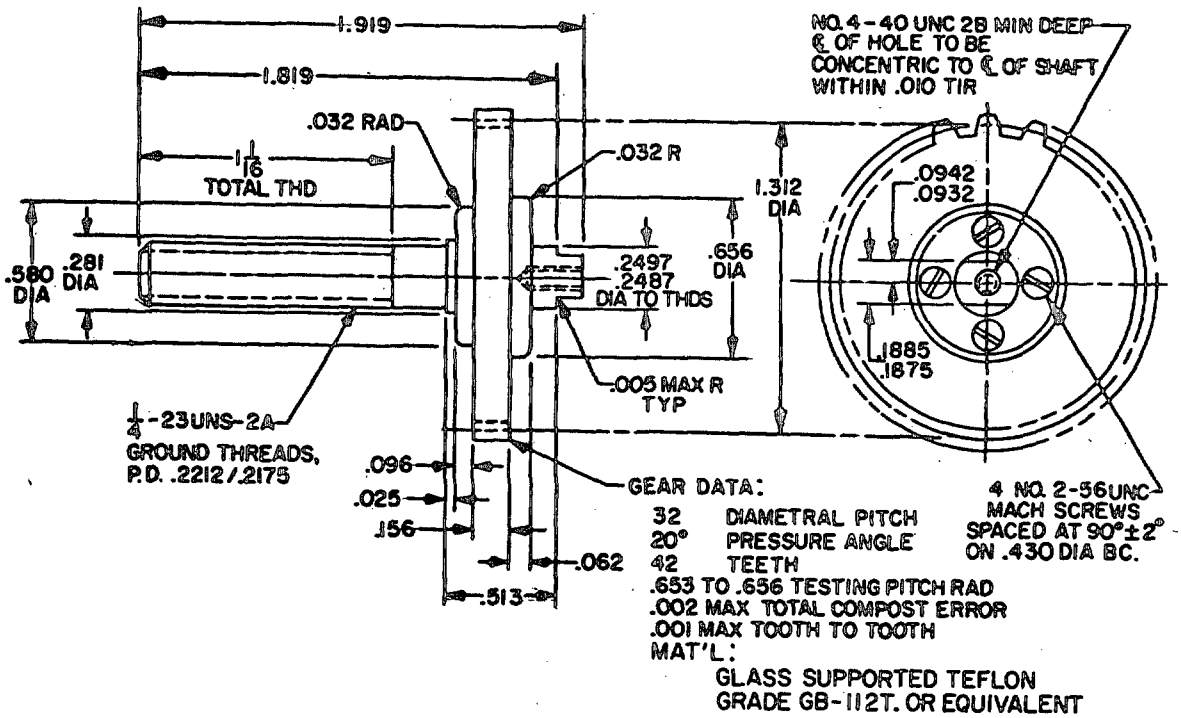


FIGURE 1c. Lead screw detail for CG66E050-471G.

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.063	1.59	7/32	5.56	.580	14.73	1.819	46.20
.002	.05	.0932	2.37	.2175	5.52	.653	16.59	1.919	48.74
.005	.13	.0942	2.39	.2212	5.62	.656	16.66	2 1/32	51.59
.010	.25	.096	2.44	.2487	6.32	.937	23.81	2.047	51.99
.015	.38	.100	2.54	.2497	6.34	.990	25.15	2.063	52.39
.025	.64	.121	3.07	1/4	6.35	1.010	25.65	2.281	57.94
.030	.76	.156	3.96	.265	6.73	1 1/16	26.97	2.313	58.74
.031	.79	.175	4.44	.281	7.14	1.156	29.37	2.344	59.53
.032	.81	.1875	4.76	.430	10.92	1 3/16	30.16	3.375	85.73
.045	1.14	.1885	4.79	.437	11.11	1.312	33.32	3.500	88.90
.062	1.57	.202	5.13	.513	13.03	1.313	33.34	5.094	129.38

NOTES:
1. Eccentricity between dia marked * not to exceed .002 TIR.
2. Runout of surface indicated measured at .281 Rad not to exceed .010 TIR with respect to diameter marked *.
3. Part to be free of burrs.
4. Centers optional.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style CG65 and CG66 capacitors - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration - See figure 1.

Envelope - Glass or ceramic.

Peak rated voltage: At:

Minimum capacity - 10 kV.

Maximum capacity - 5 kV.

Rated current (rms) - 40 amperes.

Operating temperature range - -55°C to +100°C.

(B) Capacitance:

Minimum capacity: No more than 5 pF at 16 1/4 turns from the shorted position.

Maximum capacity: No less than 465 pF at 1/4 turn from the shorted position.

Capacitor shall be short circuited at maximum capacity (end of travel).

(B) Mechanical adjustments:

Torque:

End stops - Not applicable.

Tuning shaft - Shall not exceed 40 inch-ounces when applied through a 1/4-23 lead screw from 10 pF through maximum capacitance position.

Capacitance change vs rotation:

Linear ±10 percent from 50 pF to maximum.

Vibration, high frequency: Method 204 of MIL-STD-202:

Test condition - A.

Duration and direction - 4 1/2 hours in each of 2 mutually perpendicular planes, one of which shall be axial (total of 9 hours).

Measurements during test - During the last 30 minutes of each direction, a signal shall be placed across the capacitor and measured with a suitable recorder to determine the presence of open or short circuits, or intermittent conditions.

(B) Rated load: In accordance with MIL-C-23183; test frequency shall be 16 MHz.

Type designation:

Without gear - CG65E050-471G.

With gear - CG66E050-471G.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MU, EL
Navy - EC
Air Force - 11, 17, 80, 85
DSA - ES

User activities:
Army - None
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1204)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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